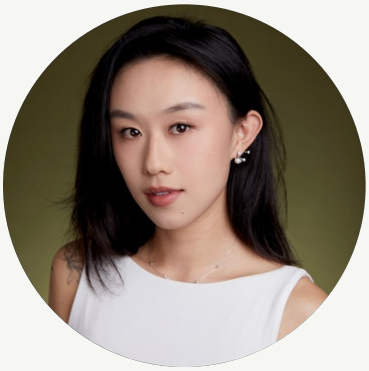
RESEARCH PORTFOLIO

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Human-AI collaboration in teams and organizations

Teams and organizations accomplish complex work by coordinating expertise, interpreting incomplete information, and evaluating contributions whose relevance depends on what others know and do. AI systems designed around an individual user can assist with idea generation or task completion, but these uses leave open how assistance changes the collaboration through which collective outcomes emerge.

The research presented here examines that relationship across entrepreneurship, science, and knowledge work. We investigate how AI can scaffold the articulation of tacit reasoning, make interaction processes observable at scale, and change the distribution of ideas and judgments across a collective. These questions connect the capabilities of individual systems to the organizational practices through which people use them together.

Our approach combines human-computer interaction, design, and computational social science. Field observation identifies what collaborators need to notice, articulate, and interpret; computational analysis examines patterns across interactions and collections of outputs; system design allows us to study how assistance changes the work in practice. Each method addresses a different source of uncertainty about what support is needed and how its consequences should be evaluated.

Across the projects, AI surfaces possible risks, structures behavioral evidence, or produces contributions that humans must interpret in context. The resulting opportunities depend on the quality of that context, the validity of the measurements, and the authority people retain over how the system guides their work. The portfolio develops these conditions through completed studies, exploratory deployments, and ongoing empirical research.

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RESEARCH OVERVIEW

AI in collective work

Supporting collective work requires understanding how assistance changes what people articulate, how they interact, and which contributions shape subsequent decisions.

Making expertise usable

Entrepreneurs seeking assistance must identify what they do not know, locate relevant experience, and explain their needs in a form that another person can address. Our coaching studies examine how mentors scaffold these activities across a learning community, then translate an empirical model of those practices into an AI system. The system prepares novices and mentors differently for their shared conversation, while allowing mentors to revise the knowledge that guides its questions. Related studies of workplace learning and literary judgment examine the capabilities and representations that make tacit knowledge available for use. Pages 3-5.

Understanding collaboration

Team processes unfold through sequences of contributions, responses, and adjustments that aggregate outcomes cannot fully explain. We develop a methodological agenda for using language, audio, and visible behavior to study those sequences at greater scale and temporal resolution. Ongoing projects in scientific teams, entrepreneur-investor exchanges, and professional volleyball examine how task demands shape knowledge integration, evaluation, and coordination. These applications require construct-specific validation before behavioral traces can support interpretations of collective work. Pages 6-7.

Evaluating collective outcomes

A collection of individually plausible AI outputs may still contain redundant ideas or similar evaluative perspectives. Our work on scientific proposals and reviews examines diversity across generation and evaluation, distinguishing the properties of a collection from the quality of any single contribution. Research on harassment tools adds a question about influence: the people supplying information may have limited authority over how it is classified or acted upon. Together, these projects examine how computational systems distribute knowledge, judgment, and decision authority. Pages 8-9.

The three streams provide complementary foundations for intervention. Coaching research specifies the human practices that can guide AI behavior; multimodal research develops measures of how collaboration unfolds; collective-outcomes research identifies consequences that individual performance measures may miss. Connecting these foundations makes it possible to ask whether assistance grounded in empirical models of collaboration produces better collective outcomes, while preserving differences in context and perspective.

The portfolio closes with research and applied experience, including a system that turns organizational expertise into shared, revisable workflows. This work provides a practical setting in which to examine who specifies common standards, how those standards adapt to unfamiliar tasks, and how users can inspect the reasoning embedded in shared support. Page 10.

STREAM 01 / CSCW 2023

Making expertise usable

Coaches coordinate learning by translating experience across projects, locating relevant expertise, and helping novices formulate requests that others can address.

We examined a university incubator in which coaches supported several novice entrepreneurs working on different ventures, stages, and timelines. Through interaction analysis of 24 virtual coaching sessions involving three coaches and 30 novices, supplemented by retrospective coach interviews and Slack observations, we developed a cognitive model of one-to-many coaching. The recordings covered 14 hours across eight weeks; the novices formed a learning community rather than necessarily belonging to one venture team. [1]

Our analysis identified several practices through which coaches made distributed experience usable. Coaches questioned assumptions and missing evidence, translated project-specific experiences into transferable lessons, and connected novices to peers with relevant expertise. They also helped novices articulate what assistance they needed and why, making it easier for others to recognize where they could contribute. For example, an account of a venture's particular testing problem could become a lesson about how to design an experiment that other entrepreneurs could apply.

Coaches also organized the conditions under which novices experimented and learned from one another. Supportive peer testing and reframing setbacks helped novices engage with uncertain results, while public commitments provided accountability. Visibility could also reproduce weak norms when vague goals went uncorrected. These findings suggest that intelligent coaching systems should attend to the practices that make knowledge transferable and requests actionable, alongside the information available about each project. The study provides a qualitative account of one incubator; it does not establish causal effects on venture performance or psychological safety.

The contribution is an empirical account of the work required to make a learning community's knowledge usable. Accounts of novice challenges primarily came from observations and coaches' interpretations rather than novice interviews, limiting conclusions about novices' own experiences. The cognitive model informs the subsequent AI system, while the effectiveness of that intervention requires evidence from its own deployment.

REFERENCES

[1] Huang, E., Rees Lewis, D., Gaudani, S., Easterday, M., & Gerber, E. (2023). Intelligent Coaching Systems: Understanding One-to-many Coaching for Ill-defined Problem Solving. Proceedings of the ACM on Human-Computer Interaction, 7(CSCW1). doi:10.1145/3579614.

STREAM 01 / CSCW 2025

Scaffolding coaching preparation

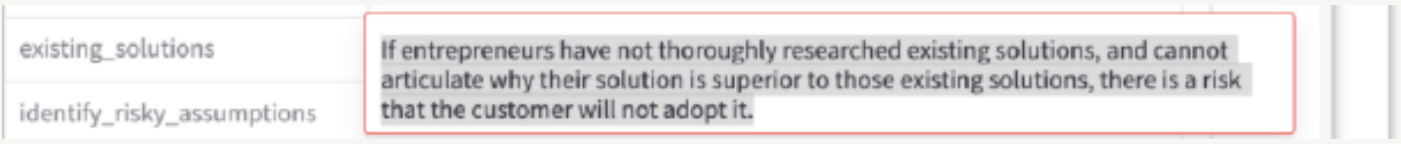
The system uses an explicit coaching model to diagnose possible risks and organize context, while mentors interpret that context and revise the model.

Preparing a coaching meeting requires novices to externalize reasoning that mentors cannot directly observe, while mentors must interpret sparse updates and plan assistance under limited time. We translated the cognitive coaching model into a proactive AI system that scaffolds these complementary activities. The system combines an LLM with project and risk models, a reflective chatbot, dashboards, and an authoring interface through which mentors can revise the knowledge guiding AI behavior. [2]

For novices, the system uses project context to diagnose possible risks, generate reflective questions, and support the selection of a meeting agenda. For mentors, it presents project updates, selected and omitted risks, and suggestions informed by mentor-specified coaching goals. Mentors can inspect the underlying responses and revise risk definitions when the existing model misses an important concern. This division of labor makes the system's support responsive to both participants, while leaving the interpretation of incomplete or ambiguous context with the mentor.

We iterated prototypes with four mentors and four novices before conducting an exploratory deployment with a different group of one mentor and 11 novices, each preparing for one actual coaching meeting. Interviews, think-aloud sessions, surveys, recordings, and comparisons of articulated risks and plans provided situated evidence of how the system shaped preparation. Novices reported identifying neglected risks and reconsidering tasks that did not address critical uncertainties; participants described subsequent conversations as more focused, intentional, and in-depth.

The deployment also exposed constraints on this support. Incomplete novice input could produce misleading diagnoses and additional verification work, and expectations of immediate answers sometimes conflicted with reflective questioning. These findings inform the design of context-sensitive preparation, but do not establish long-term skill transfer, venture success, or superiority to a generic chatbot. Future evaluations should examine whether repeated scaffolding develops collaborators' own diagnostic judgment and when it may create dependency.



Interface detail: a mentor can revise the risk definition guiding subsequent AI questions. Excerpt from Huang et al. (2025), Fig. 4B [2]; original highlighting retained.

REFERENCES

[2] Huang, E., Easterday, M., & Gerber, E. (2025). AI That Helps Us Help Each Other: A Proactive System for Scaffolding Mentor-Novice Collaboration in Entrepreneurship Coaching. Proceedings of the ACM on Human-Computer Interaction, 9(7). doi:10.1145/3757549. Open-access paper / System code.

STREAM 01 / RELATED FOUNDATIONS

Learning and tacit judgment

Making knowledge visible creates a resource for collaboration, but people also need opportunities to develop the capabilities to interpret and use it.

Learning during complex work [3]

Complex work requires people to articulate requests, seek and provide assistance, and coordinate expertise as their understanding of a problem changes. Access to colleagues and communication tools creates opportunities for collaboration, but developing the skills to use those opportunities requires support within everyday work. With an interdisciplinary group of collaborators, we developed a workshop proposal and research agenda examining how organizational structures, routines, and technologies can scaffold that learning.

This agenda connects the immediate value of AI-supported preparation to the capabilities people develop through repeated practice. For example, a system may help a novice identify an overlooked risk before one meeting, while the novice's ability to recognize similar risks independently remains uncertain. Evaluation should therefore distinguish assistance with the current task from changes in what people learn to notice, articulate, and coordinate over time. The paper proposes a workshop and research agenda; it does not report completed workshop outcomes or an intervention trial.

Representing tacit literary judgments [4]

Literary style provides an earlier setting for studying judgments that people can recognize but struggle to describe formally. With Sarah Sterman, Vivian Liu, and Eric Paulos, we conducted formative interviews with 14 participants, elicited crowdsourced comparisons, and developed a neural model of stylistic similarity. Participants' use of comparisons and examples informed a representation that did not require them to define stylistic categories in advance.

Explorer and Editor interfaces made that representation available for reflection on patterns within and across texts. In an exploratory study with six users, the visualizations prompted participants to articulate distinctions and question assumptions, although model and projection errors could also create confusion. Some users found the overall pattern more useful than exact positions or colors, suggesting that interpretation depended on how the representation was read as well as what it encoded.

The model captured a partial shared signal in judgments with substantial disagreement; it did not establish a universal ground truth for style or improved writing and collaboration. This work informs an approach carried into later projects: elicit tacit reasoning, make selected aspects inspectable, and preserve human interpretation where the computational representation loses context or nuance.

REFERENCES

[3] Huang, E., Garg, K., Gómez-Zará, D., Hui, J., Kulkarni, C., Massimi, M., Churchill, E. F., & Gerber, E. (2023). Supporting Workers in Developing Effective Collaboration Skills for Complex Work. CSCW Companion, 486-489. doi:10.1145/3584931.3611290.

[4] Sterman, S., Huang, E., Liu, V., & Paulos, E. (2020). Interacting with Literary Style through Computational Tools. CHI. doi:10.1145/3313831.3376730.

STREAM 02 / JOD PERSPECTIVE

Measuring team interaction

Multimodal AI could make collaboration observable across settings and over time, provided that measurements remain valid for the constructs and contexts being studied.

With Brian Uzzi and Matt Groh, we develop a methodological agenda organized around three dimensions: scale, resolution, and temporal continuity. Linking language, vocal cues, and visible behavior could allow researchers to compare patterns across settings, connect organizational constructs to enacted behavior, and trace how collaboration changes over an interaction. [5]

Each dimension addresses a different limitation of current observation. Scale creates opportunities to test when behavioral relationships generalize; resolution makes responses to disagreement, mistakes, and requests for help available for analysis; temporal continuity allows researchers to examine escalation, repair, and turning points. For example, comparing reported psychological safety with observed responses to dissent would make the relationship between perceptions and practice an empirical question.

These opportunities depend on construct-specific validation. Performance on a general video benchmark does not establish that a model reliably measures knowledge integration or cohesion in organizational settings. Our empirical program develops theory-grounded codebooks, compares AI annotations with human judgments, examines errors, and relates interaction patterns to collective outcomes. Organizations using such measurements should also establish who can define, inspect, and contest them, because behavioral visibility can otherwise become surveillance or encourage people to optimize narrow signals of performance.

The article presents a perspective and research agenda; it does not report validation or outcome results from the empirical projects below. Those projects use AI to study human interaction and provide a potential foundation for later interventions in which AI participates in collaboration.

The proposed capability concerns process explanation. A behavioral record could locate when a contribution is taken up, when disagreement escalates, or when a team repairs a breakdown, allowing researchers to examine sequences that an average participation score would obscure. Designing feedback from those records also requires decisions about what teams need to interpret and which inferences the evidence can support.

REFERENCES

[5] Huang, E., Uzzi, B., & Groh, M. (2026). Measuring What Matters in Teams with Multimodal AI. Journal of Organization Design. To appear.

STREAM 02 / ONGOING STUDIES

Interaction across settings

Scientific discussion, investor evaluation, and professional sport create different demands for integrating knowledge, responding to challenge, and coordinating interdependent action.

Scientific teams

Scientific collaboration requires members to evaluate and connect contributions whose relevance may become apparent only through discussion. We examine integration behavior: how members build on, connect, evaluate, and extend one another's ideas. Current analyses suggest that effective teamwork is distinguished in part by what the team subsequently does with its members' contributions. The manuscript in preparation with Max Chalekson, Matt Groh, D. Abrams, and Brian Uzzi examines these relationships as a foundation for later support that scaffolds integration.

Entrepreneur-investor interaction

Questions and challenges require entrepreneurs to clarify assumptions, substantiate claims, and respond to concerns that may not appear in a prepared presentation. With Ava Grey, Brian Uzzi, and Matt Groh, we study how the founder-evaluator exchange reveals preparedness, develops rapport, and relates to evaluators' judgments. A codebook guides annotation of interaction events; human coding provides a basis for examining agreement; and structured records preserve who contributed, when, and how a response connects to an earlier question. Findings about evaluation outcomes remain to be established.

Professional volleyball

Professional volleyball requires tightly interdependent action, with limited time to coordinate between periods of play. We use team huddles to examine how cohesion is enacted through observable interaction and how those dynamics relate to subsequent performance. The study examines behavioral relationships; effects of interventions intended to improve teamwork require separate evaluation.

Across these settings, the program draws on more than 1,000 hours of recorded interaction. Their availability does not imply that all have been annotated or that every construct is validated. Comparing settings allows us to examine which patterns depend on task uncertainty, interdependence, or evaluative pressure, establishing boundaries that context-sensitive AI support would need to respect.



Measurement sequence for the entrepreneurship study. Agreement with human coding is examined separately; the diagram describes the method and does not report validation results.

STREAM 03 / WORKING MANUSCRIPT

AI and collective outcomes

Evaluating scientific AI requires examining the range of possibilities and judgments it contributes to a decision process, alongside the quality of individual outputs.

With Matt Groh and Brian Uzzi, we compare 23 proposals and 85 expert reviews from a real bioscience grant competition with outputs generated by Claude, Gemini, and GPT under the same call and evaluation criteria. The computational experiments vary batch generation, independent calls, and calls conditioned on scientist personas. Biomedical text embeddings, diversity and geometric-coverage measures, topic modeling, matched set sizes, and style-normalized comparisons allow us to examine the distributions of proposals and reviews.

The working manuscript identifies two forms of compression. During generation, AI proposals cover broadly similar scientific territory while concentrating around common ideas, leaving some distinctive human proposals underrepresented. Distinctiveness is defined relative to the computational representation and proposal set; it does not establish that an underrepresented proposal is more novel or scientifically valuable.

During evaluation, AI panels can cover much of the semantic territory of human reviews while producing more similar judgments, particularly in their criticisms. Coverage and diversity therefore capture different properties: a panel may address familiar topics without preserving the same range of reasons for questioning a proposal. This distinction suggests that expanding the number of AI-generated reviews may not by itself expand the perspectives available to a decision process.

Several findings persist across original and style-normalized text, although effects vary by model, prompting condition, and measure. Independent calls and persona conditioning do not restore human-level semantic diversity in the tested conditions. These comparisons motivate evaluating AI by its contribution to collective search and evaluation, while limiting the conclusion to the models, prompts, and representations studied.

The study compares human and AI outputs from one competition. It does not test how scientists work with AI assistance, an end-to-end automated funding process, or realized innovation. Future evaluations should examine whether interventions that broaden candidate ideas also preserve independent sources of judgment during selection, since the two forms of compression create different design requirements.

This distinction matters because idea generation and evaluation allocate attention in different ways: one determines which possibilities enter consideration, while the other shapes the reasons for pursuing or rejecting them. The analysis treats their relationship as a question for further study, rather than demonstrating how the two compressions interact in an operational funding process.

STREAM 03 / DIS 2024

Participation and influence

Computational interventions distribute authority through decisions about whose experiences matter, which categories apply, and what responses a system makes available.

Computational approaches to online harassment depend on decisions about whose experiences matter, what counts as harm, and which responses a system makes available. Through a scoping review and qualitative analysis of 17 papers, we examined how interventions reflected intended beneficiaries' identities, definitions of harassment, and preferred forms of recourse. [6]

Our analysis identified gaps in the documented connection between beneficiaries' needs and implemented systems. Papers could motivate a tool through disproportionate harm to marginalized groups without explaining how its features addressed those groups' specific circumstances. Users could also report an experience while platforms retained authority over whether it counted as harassment and what response followed. Participation in supplying information therefore did not necessarily entail influence over classification or action.

The review identified five response strategies: punishment, distancing from perpetrators, educating perpetrators, documenting and reporting evidence, and emotional or social support. Configurable tools and community-based moderation offered ways to bring decisions closer to users, but meaningful influence depended on whether people could understand and use those options. Some gaps may reflect reporting rather than demonstrated system failures, so the findings concern documented alignment rather than a causal comparison of effectiveness.

This work informs our broader analysis of collective AI by making the allocation of interpretive and decision authority explicit. Systems that collect user input should specify how that input can affect categories, judgments, and available actions. In the wider research program, this question connects mentors' ability to revise a coaching model with teams' ability to interpret behavioral measurements and researchers' ability to preserve different evaluative perspectives.

The review contributes an account of how human-centered objectives can become disconnected from implementation and evaluation. Its implications concern the relation between participation and consequential decisions: collecting more information cannot resolve that disconnection unless the system specifies who can interpret the information and how it can change the response. The paper received an Honorable Mention at DIS 2024.

REFERENCES

[6] Huang, E., Sarma, A., Hwang, S., Chandrasekharan, E., & Chancellor, S. (2024). Opportunities, Tensions, and Challenges in Computational Approaches to Addressing Online Harassment. DIS, 1483-1498. doi:10.1145/3643834.3661623.

RESEARCH AND APPLIED EXPERIENCE

Research, design, and practice

Field partnerships and applied system development provide settings for examining how organizational knowledge becomes visible, usable, and open to revision.

Research at Northwestern

I am a Research Assistant Professor at Northwestern University's Kellogg School of Management (July 2026-present), following postdoctoral work with Brian Uzzi and Matthew Groh (February 2025-June 2026). My research connects multimodal studies of team interaction with computational experiments on AI in scientific knowledge production. I independently wrote the proposal for Building Behavioral Process Models from Real Teams with Multimodal AI, which received $75,000 from Microsoft's AI and the New Future of Work program, with Brian Uzzi as PI. The related TeamLens proposal was a finalist in the Stanford HAI and Google DeepMind AI for Organizations Grand Challenge.

Shared AI workflows at Novohaven

Since October 2025, I have worked as an AI Advisor and Forward Deployment Engineer, designing and developing a platform that makes company practices available through shared workflows. In an international e-commerce business, the structure and quality of AI-assisted deliverables depended on how employees articulated requests and supplied company context. The platform specifies task steps, information requirements, and company standards, with execution histories that allow users to inspect where revision is needed.

Domain experts can describe and revise workflows through natural-language instructions or direct specification of individual steps. The system is designed to surface recurring revision requests, while proposed changes require human review before becoming shared practice. This arrangement assigns the system a role in identifying repeated problems and retains human responsibility for changes to common standards. Effects on learning, quality, and coordination require dedicated empirical evaluation.

Education and other applied work

My PhD in Technology and Social Behavior at Northwestern (2024) bridged computer science and communication studies. A sustained partnership with an entrepreneurship incubator connected field observation, cognitive modeling, and system deployment; I also coached engineering students in human-centered design. I earned an MS in the same program (2023) and a BA in Economics from UC Berkeley (2017).

As an AI Technical Advisor to Spark AI (December 2024-December 2025), I advised product and engineering teams on AI features, integration, and deployment for K-12 learning products. At Northwestern Memorial Hospital (summer 2022), I studied older adults' medication and scheduling workflows in electronic health records and translated usability findings into design recommendations.

Outside work, I play guitar and tennis, take film and digital photographs, read, doodle, and cook spicy food. To discuss related research: jiaxinh.evey@gmail.com.